

1 The sign test

Assume X is a continuous random variable with unknown median m . Upon taking a random sample $X_1, \dots, X_n$, we'll be interested in testing whether the median m takes on a particular value m_0 . That is, we'll be interested in testing the null hypothesis:

$$H_0 : m = m_0$$

against any of the possible alternative hypotheses:

$$H_1 : m \neq m_0 \quad \text{or} \quad H_1 : m > m_0 \quad \text{or} \quad H_1 : m < m_0$$

Let's start by considering the quantities $X_i - m_0$ for $i = 1, \dots, n$. If the null hypothesis is true, that is, $m = m_0$, then we should expect about half of the $x_i - m_0$ quantities obtained to be positive and half to be negative:

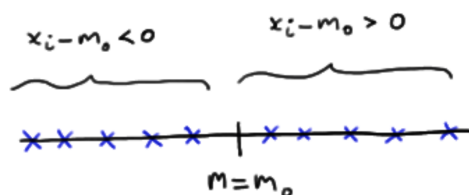


Figure 1: Expected behaviour when $m = m_0$

If instead, $m > m_0$, then we should expect more than half of the $x_i - m_0$ quantities obtained to be positive and fewer than half to be negative. If this is the case, the probability p of finding a value below m_0 is < 0.5 . This situation is illustrated in Figure 2.

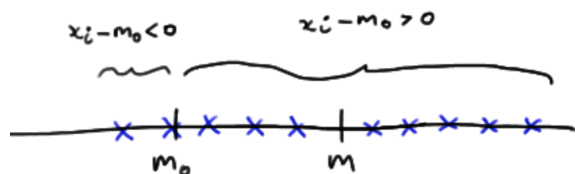


Figure 2: Expected behaviour when $m > m_0$

Or, if instead, $m < m_0$, then we should expect fewer than half of the $x_i - m_0$ quantities obtained to be positive and more than half to be negative. Then, the probability of finding an observation below m_0 is > 0.5 . This situation is illustrated in Figure 3.

This analysis of $x_i - m_0$ under the three situations $m = m_0$, $m > m_0$ and $m < m_0$ suggests then that a reasonable test for testing the value of a median m should depend on $X_i - m_0$. That's exactly what the sign test for a median does. This is what we'll do:

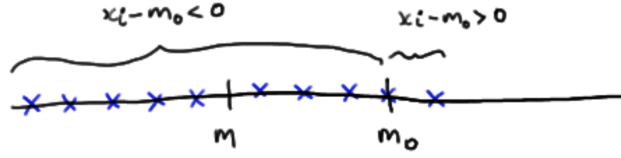


Figure 3: Expected behaviour when $m < m_0$

1. Calculate $X_i - m_0$ for $i = 1, \dots, n$.
2. Define $N-$ as the number of negative signs obtained upon calculating $X_i - m_0$ for $i = 1, \dots, n$.
3. Define $N+$ as the number of positive signs obtained upon calculating $X_i - m_0$ for $i = 1, \dots, n$.

Then, if the null hypothesis is true, $m = m_0$, then $N+$ and $N-$ both follow a binomial distribution with parameters n and $p = 0.5$. That is:

$$N+ \sim B(n, 0.5) \quad N- \sim B(n, 0.5)$$

1.1 Unilateral tests

If we are interested in testing the null hypothesis $H_0 : m = m_0$ against the alternative $H_1 : m > m_0$, then, if the alternative hypothesis were true, we should expect $X_i - m_0$ to yield more positive signs that would be expected if the null hypothesis were true (see Figure 2). In this case, we should reject the null hypothesis if $n+$, the observed number of positive signs is too large, or alternatively, if the p-value define as:

$$p - value = P(N+ \geq n+)$$

is less than or equal to the significance level α . The following three tests are equivalent:

$$\left. \begin{array}{l} H_0 : m = m_0 \\ H_1 : m > m_0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : m \leq m_0 \\ H_1 : m > m_0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : p \leq 0.5 \\ H_1 : p > 0.5 \end{array} \right\}$$

where p is the proportion of positive values in $x_i - m_0$ or the proportion of values greater than m_0 .

If we are interested in testing the null hypothesis $H_0 : m = m_0$ against the alternative $H_1 : m < m_0$, then, if the alternative hypothesis were true, we should expect $X_i - m_0$ to yield more negative signs that would be expected if the null hypothesis were true (see Figure 3). In that case, we should reject the null hypothesis if $n+$, the observed number of positive signs, is too small, or alternatively, if the p-value as defined by:

$$p - value = P(N+ \leq n+)$$

is less than or equal to the significance level α . The following three tests are equivalent:

$$\left. \begin{array}{l} H_0 : m = m_0 \\ H_1 : m < m_0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : m \geq m_0 \\ H_1 : m < m_0 \end{array} \right\} \Leftrightarrow \left. \begin{array}{l} H_0 : p \geq 0.5 \\ H_1 : p < 0.5 \end{array} \right\}$$

where p is the proportion of positive values in $x_i - m_0$ or the proportion of values larger than m_0 (see Figure 3).

1.2 Bilateral tests

If we are interested in testing the null hypothesis, $H_0 : m = m_0$, against the alternative hypothesis $H_1 : m \neq m_0$, it makes sense that we should reject the null hypothesis if we have too few negative signs or too few positive signs. Formally, we reject if n_{min} , which is defined as the smaller of $n-$ and $n+$, is too small. Alternatively, we reject if the p -value as defined by:

$$p - value = 2P(N_{min} \leq \min(n-, n+))$$

is small, that is, less than or equal to α .

The sign test can also be used with **paired data** to test, for instance, if a treatment is effective. In this case, the sign test will use the difference between the treatments response and will test whether the median of such differences is 0. The sign test will use only the information provided by the signs of such differences. The only thing needed is to evaluate which observation in a matched pair has the better response. For instance, imagine that we are comparing two tanning methods. We can choose a crossover design in which the participants get a tan using one method, and after it wears off they get a tan using the other method. The order of using the two methods is random. For each participant, we observe which method gives the better tan. That is, we make comparisons by pairing the two observations for the same participant. The only information collected about each participant is which method had better results, the first or the second. As the sign test requires only this kind of information, this test is appropriate in these situations. As a second example, imagine we want to know whether students spend more time browsing the internet or watching TV. If the only information we have about each participant is the activity they spend more time on, then, the sign test is appropriate. But if we record the actual time they spend on each activity per week, a test that uses the actual values will be more convenient (for instance, the Wilcoxon signed-rank test).

Example 1 In a crossover design to compare two different treatments to relieve headache, participants are randomly assign to one of them. After a reasonable time, they would follow the other treatment and record which one had a better result. In a sample of $n = 200$ subjects, 120 subjects found the second treatment was better to relieve headaches. Do these data provide evidence on which treatment is better?

For each subject, two responses are collected about the effectiveness of each treatment. These responses are 1 and 0, depending on which treatment was better. The structure of our data is the following:

Subject	Treat. 1	Treat. 2
1	1	0
2	0	1
$\vdots$	$\vdots$	$\vdots$
200	0	1

Difference between treatments would be positive if the first one was better and negative otherwise. In our sample, there are 120 negative signs and 80 positive signs. In testing the null that the median of these data is 0 and no treatment is better ($H_0 : m = 0$) against the alternative that the median is less than 0 and thus it is better the second treatment ($H_1 : m < 0$), the p-value is:

$$p - value = P(N+ \leq n+) = P(N+ \leq 80) = 0.002842$$

In this case, the null would be rejected. There is evidence to state that the second treatment is better.

Example 2 (Tanning experiment, small sample size) We want to compare two tanning methods without exposure to the sun to avoid skin cancer risk. We want to compare two treatments—a bronze tanning lotion applied twice over a two-day period, or a tanning studio where the person is exposed to ultraviolet (UV) light. We'll refer to these treatments as “tanning lotion” and “tanning studio.” In a crossover design, 5 participants get a tan using one method (randomly chosen) and after it wears off they get a tan using the other method. For each participant, we observe which treatment gives the better tan. The results of the experiment were that the tanning studio gave a better tan than the tanning lotion for four of the five participants. For each participant, our data looks like this:

Subject	Lotion	Studio	Difference=L-S
1	0	1	-1
2	0	1	-1
3	0	1	-1
4	0	1	-1
5	1	0	1

To explore the differences between the two methods, we get 4 negative signs and one positive sign. To test whether the tanning studio gives better results, we perform the sign test with $H_1 : m < 0$. The p-value is:

$$p - value = P(N+ \leq 1) = 0.187, \quad N+ \sim Bin(5, p = 0.5)$$

In summary, the evidence is not strong that more participants get a better tan from the tanning studio than the tanning lotion. With such a small sample size, the smallest p-value that we could obtain is $P(N+ = 0) = 0.03125$ when all the participants get a better result with the tanning studio. Note that the p-value is the second smallest value that can be obtained with this sample size. To get more interpretable p-values, we should increase the sample size.

Example 3 (paired data; using the sign of the differences) A new medicine against hypertension was tested on 18 patients. After 40 days of treatment, the following changes (after – before) of the diastolic blood pressure were observed.

–5, –1, +2, +8, –25, +1, +5, –12, –16 –9, –8, –18, –5, –22, +4, –21, –15, –11

Use the sign test to determine if the medicine has an effect on blood pressure. What is the p-value?

The sign test only considers the sign of the differences. The number of negative differences is 13 and the number of positive differences is 5. For the alternative hypothesis that there is a change in the hypertension, the p-value is, being $N+ \sim \text{Bin}(18, p = 0.5)$:

$$p - \text{value} = 2 \times P(N+ \leq 5) = 2 \times 0.04812 = 0.096$$

Using a significance level of $\alpha = 0.05$ we fail to reject the null hypothesis, we fail to reject that the median of the differences is 0. Thus, it seems that the new medicine has no effect on the blood pressure.

For the alternative hypothesis that the median of the differences is lower than 0 and thus the probability of getting a negative difference is greater than 0.5 and the new medicine seems to decrease the blood pressure, the p-value is:

$$p - \text{value} = P(N+ \leq 5) = 0.04812$$

The p-value is very close to 0.05, the reference level. As it is lower, we have evidence (very little evidence) to reject the null and state that the median of the differences is lower than 0 and the probability of getting a negative difference is greater than 0.5. A negative difference means the blood pressure after is lower than the blood pressure before taking the medicine.

Our suggestion here is to increase the sample size to get more conclusive p-values or to perform a different test where the magnitude of the differences and not only their sign is taken into account. That is what we do in Example 4.

2 Wilcoxon signed-rank test

With matched pairs data, for each pair the sign test merely observes which treatment does better, but not how much better. The Wilcoxon signed-ranks test is a nonparametric test designed for cases in which the comparisons of the paired observations can themselves be ranked. For each matched pair of responses, it measures the difference between the responses. It tests the hypothesis,

$$H_0 : \text{population median of difference scores is } 0$$

The test uses the quantitative information provided by the n difference scores by ranking their magnitudes, in absolute value. The test statistic (let's call it W) is the sum of the ranks for the differences that are positive. It requires an extra assumption on the distribution of the data: the differences of observations have a symmetric population distribution.

The distribution of the statistic W depends on the sample size n and on the number of possible ways to assign a $+$ sign or a $-$ sign to the first n integers. The mean and variance for such distribution are:

$$E(W) = \frac{n(n+1)}{4}, \quad Var(W) = \frac{n(n+1)(2n+1)}{24}, \text{ if } H_0 \text{ is true}$$

For large n , a Normal distribution can be used to calculate the p-value.

Example 4 (paired data; using the magnitude of the differences) The same data as in Example 3 before but now this test takes into account the magnitude of the differences, not only their sign. We order the differences in absolute value, but keeping their sign, and assign ranks. As there are ties, each of the 1's (positive and negative) is assigned the rank 1.5, each of the 5's (positive and negative) is assigned rank 6 (the average of 5, 6 and 7) and each of the 8's (positive and negative) rank 8.5, the average of 8 and 9. The first line in the following table represents the ordered differences and their ranks are represented in red.

-1	+1	+2	+4	-5	-5	+5	-8	+8	-9	-11	-12	-15	-16	-18	-21	-22	-25
1.5	1.5	3	4	6	6	6	8.5	8.5	10	11	12	13	14	15	16	17	18

The value of the statistic is the sum of the ranks of the positive differences (it could be also the sum of the ranks of the negative ones). Its value is $W = 1.5 + 3 + 4 + 6 + 8.5 = 23$. Let's solve the unilateral test with the alternative that the median of the differences is less than 0, $H_1 : m < 0$, and the values *after the treatment* are lower than the values *before the treatment* and the medicine decreases the blood pressure. Since there are ties in the data, the exact p-value cannot be computed. We use R to get an approximate p-value based on the Normal distribution but we can do it manually as well. With $n = 18$, the mean and standard deviation of the test statistic W are 85.5 and 22.96193 respectively. The approximate p-value, without continuity correction, is then:

$$P(W \leq 23) \approx P\left(Z \leq \frac{23.5 - 85.5}{22.96193}\right) = P(Z \leq -2.7) = 0.0034$$

This is approximately the p-value that we obtain with the following code in R:

```
wilcox.test(diff, alternative="less", exact=F, correct=T)
```

Thus, we have enough evidence to reject the null and the new medicine seems to decrease the blood pressure of the participants.

3 Wilcoxon rank sum test

The Wilcoxon non parametric test for comparing two groups tests the null hypothesis that two independent samples come from the same population distribution. We have **independent random samples** of a quantitative random variable from two groups, for instance, two treatments, and we want to test the null hypothesis that they both come from the same population distribution. Let $X_1, \dots, X_n$ denote a sample from the first population and $Y_1, \dots, Y_m$ a sample from the second. If we let F and G , both assumed to be continuous, denote the distribution

functions of the two samples, respectively, then the hypothesis we wish to test is $H_0 : F = G$. The Wilcoxon test does that by ranking or ordering the $n + m$ data, $X_1, \dots, X_n, Y_1, \dots, Y_m$. Since we are assuming that F and G are continuous, this ranking will be unique— that is, there will be no ties. In practice, as the data is discretized, ties may be present. In this case, we assign to those repeated observations the average rank of the ranks they would have had if the values had differed slightly. Then, give the smallest data value rank 1, the second smallest rank 2, . . . , and the $(n + m)$ th smallest rank $n+m$. Now, for $i = 1, \dots, n$, let

$$R_i = \text{rank of the data value } X_i$$

The rank sum test utilizes the test statistic T equal to the sum of the ranks from the first sample— that is,

$$T = \sum_{i=1}^n R_i$$

When the null hypothesis is true and so $F = G$, it follows that all $n + m$ data values come from the same distribution and thus all $(n+m)!$ possible rankings of the values $X_1, \dots, X_n, Y_1, \dots, Y_m$ are equally likely. From this it follows that choosing the n rankings of the first sample is probabilistically equivalent to randomly choosing n of the (possible rank) values $1, 2, \dots, n + m$. Using this, it can be shown that T has a mean and variance given by

$$E(T) = \frac{n(n + m + 1)}{2}, \quad \text{Var}(T) = \frac{nm(n + m + 1)}{12} \quad \text{if } H_0 \text{ is true}$$

In addition, it can be shown that when both n and m are of moderate size (both being greater than 7 should suffice) T has, under H_0 , approximately a normal distribution. Thus, we can use the Normal distribution to compute p-values for this test.

Example 5 An experiment designed to compare two treatments against corrosion yielded the following data in pieces of wire subjected to the two treatments.

Treatment 1	65.2, 67.1, 69.4, 78.2, 74, 80.3
Treatment 2	59.4, 72.1, 68.0, 66.2, 58.5

(The data represent the maximum depth of pits in units of one thousandth of an inch.) The ordered values are 58.5, 59.4, 65.2*, 66.2, 67.1*, 68, 69.4*, 72.1, 74*, 78.2*, 80.3* with an asterisk noting that the data value was from sample 1. Hence, the value of the test statistic is $T = 3 + 5 + 7 + 9 + 10 + 11 = 45$. Software implementation of this test varies as there are variations of the test definition in the literature. It is a convention that the sample considered the first is the one with the smallest sample size. You can use function `wilcox.test` in R to obtain an exact or approximate p-value, depending on the samples sizes. This function in R computes the test differently. It counts the times a value in the first sample is greater than a value in the second, i.e., it computes, for $i = 1, \dots, n$ and $j = 1, \dots, m$ the number of times that $X_i > Y_j$. With this data, this value is 24. The p-value, for a bilateral test, is 0.1255. The null hypothesis that both samples come from the same distribution cannot be rejected. This is the code to use:

```
sample_1<- c(65.2, 67.1, 69.4, 78.2, 74, 80.3)
sample_2<-c(59.4, 72.1, 68, 66.2, 58.5)
wilcox.test(sample_1, sample_2, exact=T)
```

4 The Kruskal Wallis test

This procedure tests the equality of multiple probability distributions. Therefore, it extends the previous test to k populations. Suppose there are k populations, that F_i is the distribution function of some measurable value of the elements of population i , and that we are interested in testing the null hypothesis:

$$H_0 : F_1 = F_2 \cdots = F_k$$

versus

$$H_1 : \text{not all of the } F_i \text{ are equal}$$

To test the preceding null hypothesis, suppose that independent samples are drawn from each of the k populations. Let n_i denote the size of the sample chosen from population $i, i = 1, \dots, k$ and let $N = \sum_{i=1}^k n_i$ denote the total number of data values obtained. Now, rank these N data values from the smallest to largest, and let R_i denote the sum of the ranks of the n_i data values from population $i, i = 1, \dots, k$. Now, when H_0 is true, the rank of any individual data value is equally likely to be any of the ranks $1, \dots, N$, and thus the expected value of its rank is $\frac{1+2+\dots+N}{N} = \frac{N+1}{2}$. Consequently, with $\bar{r} = \frac{N+1}{2}$, it follows when H_0 is true that the expected sum of the ranks of the n_i data values from population i is $n_i \bar{r}$. That is, when H_0 is true:

$$E(R_i) = n_i \bar{r}$$

Drawing inspiration from the goodness of fit test, let us consider the test statistic

$$T = \sum_{i=1}^k \frac{(R_i - n_i \bar{r})^2}{n_i \bar{r}}$$

and use a test that rejects the null hypothesis when T is large. A simpler expression for T is:

$$T = \frac{1}{\bar{r}} \sum_{i=1}^k \frac{R_i^2}{n_i} - N \bar{r}$$

Hence, rejecting H_0 when T is large is equivalent to rejecting H_0 when $\sum_{i=1}^k \frac{R_i^2}{n_i}$ is large. So we might as well let the test statistic be:

$$T = \sum_{i=1}^k \frac{R_i^2}{n_i}$$

This last expression is the one used in the test. To determine the appropriate α level significance test, we need the distribution of T when H_0 is true. While its exact distribution is rather complicated, we can use the result that, when H_0 is true and all n_i are at least 5, the distribution of

$$\frac{12}{N(N+1)} T - 3(N+1) \sim \chi_{k-1}^2$$

is approximately that of a chi-squared random variable with $k - 1$ degrees of freedom.

Example 6 The following data give the number of visitors to a medium size Los Angeles library on Tuesdays, Wednesdays, and Thursdays of 10 successive weeks.

Tuesday visitors	721, 660, 622, 738, 820, 707, 672, 589, 902, 688
Wednesday visitors	604, 626, 744, 802, 691, 665, 711, 715, 661, 729
Thursday visitors	642, 480, 705, 584, 713, 654, 704, 522, 683, 708

Are these data consistent with the hypothesis that the distributions of the number of visitors for the three days are identical?

Ordering the $N = 30$ data values, gives that the sum of the ranks of the three samples are $R_1 = 176$, $R_2 = 175$, $R_3 = 114$. Therefore,

$$\frac{12}{N(N+1)}T - 3(N+1) = \frac{12}{30 \cdot 31} \frac{176^2 + 175^2 + 114^2}{10} - 93 = 3.254$$

The p-value is computed as:

$$p - value = P(\chi_2^2 \geq 3.254) = 0.1965$$

and the null hypothesis that all three samples come from the same distribution can not be rejected.

5 Exercises

1. An engineering firm is involved in selecting a computer system, and the choice has been narrowed to two manufacturers. The firm submits eight problems to the two computer manufacturers and has each manufacturer measure the number of seconds required to solve the design problem with the manufacturer's software. The times for the eight design problems are given below.

Design problem	1	2	3	4	5	6	7	8
Time with computer A	15	32	17	26	42	29	12	38
Time with computer B	22	29	1	23	46	25	19	47

Determine the p-value of the sign test when testing the hypothesis that there is no difference in the distribution of the time it takes the two types of software to solve problems.

We have observed 8 values of the variable *difference between computer A and computer B times*: 4 are negative and 4 are positive. The sign test will yield a p-value equal to 1. Since the variable measured is quantitative, we can take advantage of the magnitude of the differences and carry out the Wilcoxon signed-rank test. Since there are ties (repeated values) in the observed values, an exact test wouldn't be appropriate. The Wilcoxon signed-rank test with and without the continuity correction gives high p-values. There is no evidence that the execution time is different for computers A and B.

```
> diff_computers # ties are present in the data set
[1] -7 3 16 3 -4 4 -7 -9
> wilcox.test(diff_computers, exact=F, correct=FALSE)
> wilcox.test(diff_computers, exact=F, correct=TRUE)
```

2. The published figure for the median systolic blood pressure of middle-aged men is 128. To determine if there has been any change in this value, a random sample of 100 men has been selected. Test the hypothesis that the median is equal to 128 if

- (a) 60 men have readings above 128;
- (b) 70 men have readings above 128;
- (c) 80 men have readings above 128.

In each case, determine the p-value.

- There are 60 positive signs (values greater than 128) and 40 negative signs (values less than 128). For a two sided test, $N_{min} = 40$, $N_{min} \sim \text{Bin}(100, p = 0.5)$, and the p-value is:

$$2P(N_{min} \leq 40) = 2(0.02844) = 0.05688$$

To obtain the p-value in R type: `2*pbinom(40,100,p=0.5)`

If we use a 5% significance level, since the p-value is greater, although not much greater than 0.05, we don't have enough evidence to reject H_0 .

- There are 70 positive signs and 30 negative signs. For a two sided test, $N_{min} = 30$, $N_{min} \sim \text{Bin}(100, p = 0.5)$, and the p-value is:

$$2P(N_{min} \leq 30) = 2(3.925 \times 10^{-5}) = 7.85 \times 10^{-5}$$

With such a low p-value, we have enough evidence to reject the null hypothesis and make the conclusion that the median is different from 128.

- There are 80 positive signs and 20 negative signs. Thus, the p-value is even lower than the previous one.

$$2P(N_{min} \leq 20) = 2(5.579 \times 10^{-10}) = 1.116 \times 10^{-9}$$

3. Fifteen cities, of roughly equal size, are chosen for a traffic safety study. Eight of them are randomly chosen, and in these cities a series of newspaper articles dealing with traffic safety is run over a 1-month period. The number of traffic accidents reported in the month following this campaign is as follows:

Treatment Group	19	31	39	45	47	66	74	81
Control Group	28	36	44	49	52	52	60	

When testing the hypothesis that the articles haven't had any effect, which test would be the more appropriate?

Since we have two independent samples, one from each city, the null hypothesis that the articles have not had any effect means the two samples come from the same population. Thus, a Wilcoxon rank sum test is appropriate here. The exact distribution of the test can not be computed because there are ties (in the control group). The code in R is:

```
treatment=c(19, 31, 39, 45, 47, 66, 74, 81)
control=c(28, 36, 44, 49, 52, 52, 60)
wilcox.test(treatment, control, exact=F)
```

The p-value obtained is high, p-value=0.8621. We don't have evidence of any effect of the articles.